



Numerical approximation of nonlinear and non-equilibrium model of gradient elution chromatography

Nazia Rehman, Muhammad Abid, and Shamsul Qamar 

Department of Mathematics, COMSATS University Islamabad, Islamabad, Pakistan

ABSTRACT

A nonlinear lumped kinetic model of liquid chromatography is formulated and solved numerically to theoretically investigate the effect of column overloading on gradient elution. Linear solvent strength (LSS) model is utilized for Henry's constant, non linearity coefficient and axial dispersion coefficient. A semi-discrete high-resolution finite volume scheme is extended and applied to obtain the approximate solutions of the governing model equations. The effects of changing modulator concentration are examined on the single and two-component elution. The benefits of gradient elution over isocratic elution are thoroughly discussed. The influences of optimizing free-parameters available in gradient chromatography are analyzed on the efficiency of the column and on the production of targeted components. For instance, the results obtained are used to study the effects of gradient slope, modulator concentration, solvent strength, nonlinearity coefficient, mass transfer coefficient, and axial dispersion coefficient on the concentration profiles.

KEYWORDS

Gradient elution; finite volume scheme; lumped kinetic model; mass transfer; numerical approximation

Introduction

Several mixtures can easily be separated through liquid chromatography technique at fixed operating conditions. However, the optimal resolution of complex mixtures, containing components with large variations in the affinities to the solid (stationary) phase, is mostly not achievable at constant mobile phase composition (isocratic elution). This issue can be overcome by using programmed elution techniques, where the altering operating conditions ensure the optimal resolution for early eluted components during a single analytical run, while ensuring acceptably short elution times of the last eluted components.^[1] Since its introduction in the middle of 20th century,^[2–4] mobile phase gradients have been and continue to be the most widely used programming methods in liquid chromatography.^[1] According to a pre-set program during the chromatographic run, as the concentration of a strong solvent increases, the retention times of the solutes decreases and the elution strength of the mobile phase increases.^[5–8]

It is possible to combine gradient elution techniques with flow-rate programming,^[9] column switching, temperature programs or high temperature operation,^[10,11] and two-dimensional operation^[12] to maximize the advantage of separation selectivity and to isolate complex samples as quickly as possible. Appropriate column chemistry and mobile phase components should be chosen for each gradient separation prior to fixing the gradient program. Moreover, mobile phase flow rate and column dimensions may be changed to suit the sample type and detection technique such as liquid chromatography at a later stage of system development, but the gradient profile should be re-adjusted while changing other operation

conditions. The theoretical studies for gradient elution seem to be more complex as compared to isocratic elution, since the retention behavior of the solvent in this operating condition will also differ with the change in the composition of the mobile phase. Snyder,^[13,14] who developed a commonly used theory of linear solvent strength (LSS) gradients, is credited for a fundamental contribution to the theory of gradient elution. In a LSS-gradient approach, the gradient program is modified so that the logarithms of retention factors of sample compounds reduce linearly.^[15,16] An excellent analysis of the linear solvent strength gradient method has been published by Snyder and Dolan.^[5] The linear solvent strength gradients are fairly well approximated in reverse phase chromatography, where gradient elution is often used. However, in some chromatographic techniques, especially in ion exchange chromatography, non-LSS strongly curving plots of retention factor with respect to modulator concentration have also been introduced.

To understand the chromatographic separation processes, different mathematical models available in the literature can be used.^[17–19] Each model discusses the chromatography process in its own level of complexity. The complexity of these mathematical models is based on the assumptions taken on thermodynamic behavior of the system, types of packing particles used, adsorption-desorption kinetics, as well as on the mass transfer and dispersion mechanisms. In liquid chromatography, the most commonly used models are the general rate model, the lumped kinetic model, and the equilibrium dispersive model. Several finite difference,

finite volume, and finite element methods have been used in the literature for solving chromatographic models^[20–24]

The finite volume methods (FVMs) belong to the class of discretization techniques that have proved to be very useful in approximating the solution of a wide range of systems of conservation law.^[20,24,25] Its application can be seen in fluid dynamics, electromagnetic, porous media flow, biological process models, meteorology, simulation of semi-conductor devices, and many other engineering areas that can be described in integral volume control form, governed by conservative systems.^[26,27] The high resolution finite volume methods (HR-FVMs) are one of the effective and precise computational schemes for numerically approximating hyperbolic partial differential equations that are known to express conservation laws for various quantities. These schemes start with the domain partitioning into many control volumes and afterwards estimating the integral conservation laws in all of these volumes.^[28,29] Accurate and reliable solutions on coarse grids can be obtained through these schemes. They have the ability to capture drastic changes in the solutions and prevent numerical oscillations in the predictions. These techniques have recently been applied to approximate chromatographic models for isocratic elution.^[17,20]

In this research, a non-linear lumped kinetic model is developed and numerically approximated for theoretically studying the effect of column overloading in gradient elution. In this model, the Henry's constants, nonlinearity coefficients, mass-transfer coefficients, and axial dispersion coefficients are depending on the solvent composition. The current research extends our previous study based on the linear transport model.^[30] Unlike to the previous study for the transport model (linear), the current investigation also considers axial-dispersion and nonlinearity of the model equations. Analytical solutions of the model equations cannot be derived for non-linear adsorption isotherms. For this purpose, the numerical simulations are required to correctly determine the dynamical behavior of chromatographic columns. The HR-FVM is extended and applied to obtain solutions of the governing equations. It has already been verified in our previous article that this scheme provides second-to-third order accuracy.^[20] The considered case studies illustrate changes in the elution behaviors of mixture components exposed to gradients. The influences of various parameters are analyzed on the process performance and the benefits of gradient elution over isocratic elution are discussed. The results also show, for situations which are closer to real applications, the strength and the potential of the model and numerical scheme derived.

The contents of this research article are structured in the following manner. In Section “The lumped kinetic model for gradient elution”, the nonlinear LKM for gradient elution is introduced. In Section “Numerical Method”, the proposed HR-FVM is derived for solving the considered model equations. Discussions on the numerical case studies, demonstrating the effectiveness of model and numerical scheme, are presented in Section 4. Conclusions on the basis of results discussions are presented in Section “Conclusion”.

The lumped kinetic model for gradient elution chromatography

In this section of the paper, the LKM is used to analyze the dynamical behavior in gradient elution chromatographic separation process. A simple model for the description of mass transfer kinetics inside the chromatographic column is the LKM. The model lumps the contributions of external and internal resistances of mass transport into the coefficient of mass transfer. The driving force is commenced from the deviation from equilibrium concentration. Two fundamental kinetic variables are included in this model, i.e. $K_{L,i}$ representing the coefficient of mass transfer of i th mixture component, and $D_{z,i}$ denoting the corresponding coefficient of axial dispersion of i th component. It is assumed that mass transfer and axial dispersion coefficient are function of the volume-fraction of modifying non-retained solvent. With these presumptions, the mass balance equations in the bulk fluid for the transport of N_c species are given as

$$\frac{\partial c_i}{\partial t} + u \frac{\partial c_i}{\partial z} = D_{z,i}(\phi) \frac{\partial^2 c_i}{\partial z^2} - FK_{L,i}(\phi)(q_i^* - q_i), \quad (1)$$

$$i = 1, 2, \dots, N_c.$$

In Equation (1), the number of species in the sample of mixture is denoted by the N_c . For each component i , the concentration of the solute in the bulk of fluid is denoted by c_i , the symbols q_i and q_i^* denote the non-equilibrium and equilibrium concentrations of species in the stationary phase, respectively. Furthermore, t is the time, u is the interstitial velocity, z is the axial coordinate, the phase ratio is $F = (\frac{1-\epsilon}{\epsilon})$, the external porosity is symbolized by ϵ , and the external mass transfer coefficient is denoted by $K_{L,i}$.

This model assumes that diffusion in the stagnant mobile phase and the adsorption-desorption processes are infinitely fast. Consequently, in addition to the above equation, in order to complete the model, the corresponding balance equation for the concentrations of the mixture component in the solid particles is also needed and it can be written in the kinetic equation form as

$$\frac{\partial q_i}{\partial t} = FK_{L,i}(\phi)(q_i^* - q_i), \quad i = 1, 2, \dots, N_c. \quad (2)$$

The equilibrium adsorption isotherms define equilibrium relations among the concentrations of mixture components in the mobile and stationary phases. For modeling and simulation of the chromatographic separation process, it is essential to represent experimentally determined adsorption equilibrium data by suitable mathematical equations. A number of different adsorption isotherm models have been developed in the literature which are classified as linear, Langmuir, BET, and Freundlich isotherms.^[17,31–35] The non-linear Langmuir isotherms are considered here. For a mixture of N_c components, the Langmuir isotherms are expressed as

$$q_i^* = \frac{K_{H,i}(\phi)c_i}{1 + \sum_{j=1}^{N_c} b_j(\phi)c_j}, \quad i = 1, 2, \dots, N_c \quad (3)$$

where

$$\begin{aligned} K_{H,i}(\phi) &= K_{Hr,i}e^{-\alpha\phi}, & b_i(\phi) &= b_i^{ref}e^{-\alpha\phi}, \\ K_{L,i}(\phi) &= K_{Lr,i}e^{-\gamma\phi}, & D_{z,i}(\phi) &= D_{zr,i}e^{-\zeta\phi}. \end{aligned} \quad (4)$$

Here, the i th component reference Henry coefficient is denoted by $K_{Hr,i}$. Moreover, $K_{Lr,i}$ is the reference value of the mass transfer coefficient, and b_i^{ref} quantifies the reference value of extent of non linearity associated with the isotherm for i th component of the mixture. The specific solvent strengths are denoted by α , γ , and ζ . While, ϕ is volume-fraction of the modifying non-retained solvent.

In this work, we are interested to find the numerical solution of this model for the linear case ($b_i^{ref} = 0$) as well as for the non-linear case ($b_i^{ref} \neq 0$). Thus, the suggested model is applicable for the simulation of both analytical and preparative liquid chromatography.

The mobile phase modulator concentration, $\phi(t, z)$, varies with the position and time and determines the solute's migration velocity inside the column. We ignore the dispersion of the solvent gradient relative to the dispersion of the solute, which, of course, is retained, as we presume that the modulator is un-retained. However, it is a crucial assumption that must be carefully checked prior to applying the model equations. The ideal model is used for describing ϕ in term of time and space in the column:

$$\frac{\partial \phi}{\partial t} + u \frac{\partial \phi}{\partial z} = 0, \quad (5)$$

with initial and boundary conditions

$$\phi(0, z) = \phi_0, \quad 0 \leq z \leq L, \quad (6)$$

$$\phi(t, 0) = \begin{cases} \phi_0, & \text{if } t < t_s, \\ \Phi(t - t_s), & \text{if } t_s \leq t \leq t_e, \\ \phi_e & \text{if } t > t_e \end{cases} \quad (7)$$

where, the gradient start and end time are denoted by t_s and t_e , respectively. The modulator's initial concentration is symbolized by ϕ_0 , while Φ represents the implemented profile of gradient and the final concentration of modulator is represented by the ϕ_e . For linear gradient

$$\Phi(t) = \phi_0 + \beta t, \quad \beta = \frac{\phi_e - \phi_0}{t_e - t_s}, \quad (8)$$

where β is the steepness (or slope) of the gradient.

To complete the model equations given in Equations (1) and (2), initial conditions along with boundary conditions at the left and right ends of the column must be imposed. We consider an initially equilibrated column, that is:

$$c_i(0, z) = 0, \quad q_i(0, z) = 0, \quad i = 1, 2, 3, \dots, N_c. \quad (9)$$

Moreover, in this study, we consider rectangular injections at the left end of the column.^[17]

$$c_i(t, 0) = \begin{cases} c_{i, inj}, & \text{if } 0 \leq t \leq t_{inj} \\ 0, & \text{if } t > t_{inj} \end{cases} \quad i = 1, 2, 3, \dots, N_c. \quad (10)$$

While, Neumann conditions are used at the right end of column:

$$\frac{\partial c_i}{\partial z}(t, L) = 0, \quad i = 1, 2, \dots, N_c, \quad (11)$$

where, the i th component injected concentration is denoted by the symbol $c_{i, inj}$, the time of injection is denoted by t_{inj} . Moreover, the injection becomes continuous when $t_{inj} \rightarrow \infty$ or when the simulation time is less than the injection time t_{inj} .

Numerical method

A number of numerical techniques have been proposed and applied by different researchers for the estimation of chromatographic models. In this section of the article, we derive the high-resolution flux-limiting finite volume method (HR-FVM) for solving the nonlinear LKM for gradient elution chromatography, i.e. for $b_i(\phi) \neq 0$. The behaviors of profile observed in the case studies are discussed thoroughly.^[6,19] After that, we present some numerical case studies for analyzing our numerical solutions for different values of the model parameters. The method is easily implementable, compact and simple. The following are the basic steps for implementation of the suggested numerical algorithm:

1. The computational domain is partitioned into finite number of grids along the axial-coordinate of the column.
2. The essential idea is to divide the domain into many control volumes and to approximate the integral conservation laws on each of the control volumes. The conservation equations are discretized in term of axial-coordinate.
3. The primary goal of this discretization process is to ultimately obtain a set of coupled ordinary differential equations (ODEs) in term of time-coordinate which are then solved by an ODE-solver.
4. For implementation of the algorithm, the MATLAB software is used.
5. The in-built ODE-solver RK-45 solver in MATLAB is utilized to solve the obtained ODE-system.

In addition, the method is capable of resolving sharp peaks and discontinuous fronts in the solutions accurately. The following multi-component LKM model will be used to analyze the dynamical behavior in gradient elution chromatography. Here, the numerical solution of the PDE in Equation (1) utilizes the HR-FVM, whereas the ODE given in Equation (2) is approximated numerically by a standard ODE solver.

Discretization of computational domain

For the implementation of proposed HR-FVM from Equations (1) to (4), the computational domain is partitioned into finite number of grids along the axial-coordinate of the column. Here, we consider a uniform discretization of the computational domain $[0, 1]$. The mesh intervals width and cell centers are represented by Δz and z_k ,

respectively. Let the number of grid points is expressed by $\mathcal{N}$, the coordinates $z_{k-\frac{1}{2}}$ and $z_{k+\frac{1}{2}}$ are the left and right boundaries of the interval. The mesh point z_k in the interval $\Omega_k \equiv [z_{k-\frac{1}{2}}, z_{k+\frac{1}{2}}]$ are covered by cells for $1 \leq k \leq \mathcal{N}$.

$$z_{1/2} = 0, \quad z_{k+1/2} = k\Delta z, \quad z_{N+1/2} = L, \quad (12)$$

and

$$z_k = \frac{z_{k-1/2} + z_{k+1/2}}{2}, \quad \Delta z_k = z_{k+1/2} - z_{k-1/2} = \frac{L}{\mathcal{N} + 1}. \quad (13)$$

Cell averages

Here, for the averaged initial data $w_k(0)$ in each cell are defined as

$$w_k(0) = w_k(t) = \frac{1}{\Delta z_k} \int_{\Omega_k} w(z, 0) dz, \quad k = 1, 2, \dots, \mathcal{N}. \quad (14)$$

Here, $w \in \{c_i, c_i, q_i^*\}$. Until now, we have discretized the computational domain and assigned the corresponding initial data set at $t=0$ to each mesh interval.

Integral form of the LKM

Assuming that the cell averages w_k^n are provided at the time level t^n . Then, we can easily find cell averages w_k^{n+1} at the next level of time i.e. t^{n+1} . Furthermore, the integrating Equations (1) and (2) over Ω_k yields

$$\begin{aligned} \frac{dc_{i,k}}{dt} = & -u \frac{c_{i,k+\frac{1}{2}} - c_{i,k-\frac{1}{2}}}{\Delta z} \\ & + \frac{D_{z,i}(\phi_k(t))}{\Delta z} \left[\left(\frac{\partial c_i}{\partial z} \right)_{i,k+\frac{1}{2}} - \left(\frac{\partial c_i}{\partial z} \right)_{i,k-\frac{1}{2}} \right] \\ & - FK_{L,i}(\phi_k(t))(q_i^* - q_i), \end{aligned} \quad (15)$$

$$\frac{dq_{i,k}}{dt} = K_{L,i}(\phi_k(t))(q_i^* - q_i), \quad i = 1, 2, \dots, N_c, \quad (16)$$

The solution of Equations (5) and (6) for ϕ_k is given as:

$$\phi_k(t) = \begin{cases} \phi_0, & \text{if } t - \frac{z_k}{u} < t_s, \\ \Phi\left(t - \frac{z_k}{u} - t_s\right), & \text{if } t_s \leq t - \frac{z_k}{u} \leq t_e, \\ \phi_e, & \text{if } t - \frac{z_k}{u} > t_e. \end{cases} \quad (17)$$

The differential terms in the diffusion part of Equation (15) can be approximated as

$$\left[\frac{\partial c_i}{\partial z} \right]_{k \pm \frac{1}{2}} = \left[\frac{c_{i,k \pm 1} - c_{i,k}}{\Delta z} \right]. \quad (18)$$

To calculate the convective fluxes $c_{ik \pm \frac{1}{2}}$ at the cell interface, various numerical schemes can be used.

First order numerical method

The values of concentration are approximated at the cell interfaces in Equation (15) by utilizing backward difference

formula in this method. For the concentration, approximation can be expressed as:

$$c_{i,k+\frac{1}{2}} = c_{i,k}, \quad c_{i,k-\frac{1}{2}} = c_{i,k-1}. \quad (19)$$

A first order accurate method can be generated by the above mentioned approximations.

The HR-FVM of koren

Here, the concentration values at the interface of a cell $c_{i,k+\frac{1}{2}}$ are approximated by the following flux-limiting formulas:^[29]

$$c_{i,k+\frac{1}{2}} = c_{i,k} + \frac{1}{2} \Psi(r_{i,k+\frac{1}{2}})(c_{i,k} - c_{i-1,k}), \quad (20)$$

where,

$$r_{k+\frac{1}{2}} = \frac{c_{i,k+1} - c_{i,k} + \eta}{c_{i,k} - c_{i,k-1} + \eta}. \quad (21)$$

Here, $r_{k+\frac{1}{2}}$ is the ratio of gradients of consecutive flux. Further, to avoid division by zero, we consider $\eta = 10^{-10}$. The limiting function Ψ in Equation (20) can be defined as:^[29,36]

$$\Psi[r_{k+\frac{1}{2}}] = \max \left[\left(0, \min \left[2r_{k+\frac{1}{2}}, \min \left[\frac{1}{3} + \frac{2}{3}r_{i+\frac{1}{2}}, 2 \right] \right] \right) \right]. \quad (22)$$

Analogously, the concentration $c_{i,k-\frac{1}{2}}$ could be computed by changing the index k with the index $k-1$.

The scheme strategy at boundaries of cells

The boundary intervals are not covered by the flux-limiting approximations of Equation (20) and (21). Consider the left boundary, which is subject to an inflow boundary condition. The interval face $z_{\frac{1}{2}}$ and the inflow boundary are in the same spot. Therefore, the first-order backward method is employed to compute cell-interface concentration in the boundary cells. The fluxes at the interfaces of interior cells can be computed by aforementioned second-order accurate HR-FVM. To solve this problem, the first order approximation can be used at the cell interfaces $z_{\frac{1}{2}}$ and $z_{N+\frac{1}{2}}$ in Equation (19). If $c_{i,inj}$ denotes the flux injected, then

$$c_{i,\frac{1}{2}} = c_{i,inj}, \quad c_{i,\frac{3}{2}} = c_{i,1}, \quad c_{i,N+\frac{1}{2}} = c_{i,N}. \quad (23)$$

ODE-solver

A standard ODE solver is utilized for the solution of the ODEs system in Equation (16) to (18). Further, the mathematical programming software "MATLAB" is utilized to program the numerical scheme and the built-in Matlab routine ode45 is utilized to obtain the solution for the resultant ODEs system.

Numerical temporal moments

Temporal moments can be used to better illustrate the graphical results received in the chromatogram during the chromatographic running phase. It is an efficient method for collecting basic information about liquid chromatography elution. Chromatographic effects depend on both the

retention equilibrium and kinetics mass transfer in the column, such as the profile of elution peak, efficiency of column, performance of separation, sample retention, band broadening, and other parameters. From our numerical

Table 1. Parameters for both linear and nonlinear adsorption isotherms for single component case.

Parameters	Values
Column length	$L = 10$ cm
Interstitial velocity	$u = 0.3$ cm/min
Porosity	$\epsilon = 0.4$
Axial dispersion coefficient	$D_{zr} = 0.002$ cm ² /min
Henry constant of component	$k_{Hr} = 7.0$
Mass transfer coefficient	$K_{Lr} = 10$ min ⁻¹
Non-linearity coefficient	$b^{\text{ref}} = 2.0$
Solvent strength parameter	$\alpha = 0.90$
Starting time of Gradient	$t_s = 5$ min
Ending time of Gradient	$t_e = 80$ min
Initial concentration	$\phi_o = 0.1$
Final concentration	$\phi_e = 0.95$

solutions, the simulated moments can be easily obtained. The n th temporal moment at the outlet of the chromatographic bed of length $z = L$ is obtained as.

$$M_n = \int_0^\infty c(t, z = L, t) t^n dt. \quad (24)$$

The n -th normalized temporal moment is represented as

$$\mu_n = \frac{M_n}{M_0} = \frac{\int_0^\infty c(t, z = L) t^n dt}{\int_0^\infty c(t, z = L) dt}. \quad (25)$$

Whereas the n -th central moment is

$$\mu_i^* = \frac{\int_0^\infty c(t, z = L) (t - \mu_1)^n dt}{\int_0^\infty c(t, z = L) dt}. \quad (26)$$

The elution curve is used in liquid chromatography to define the distribution of concentration with respect to time and is recognized as the curve of probability density. To define these density curves, the zeroth, first, second, and the

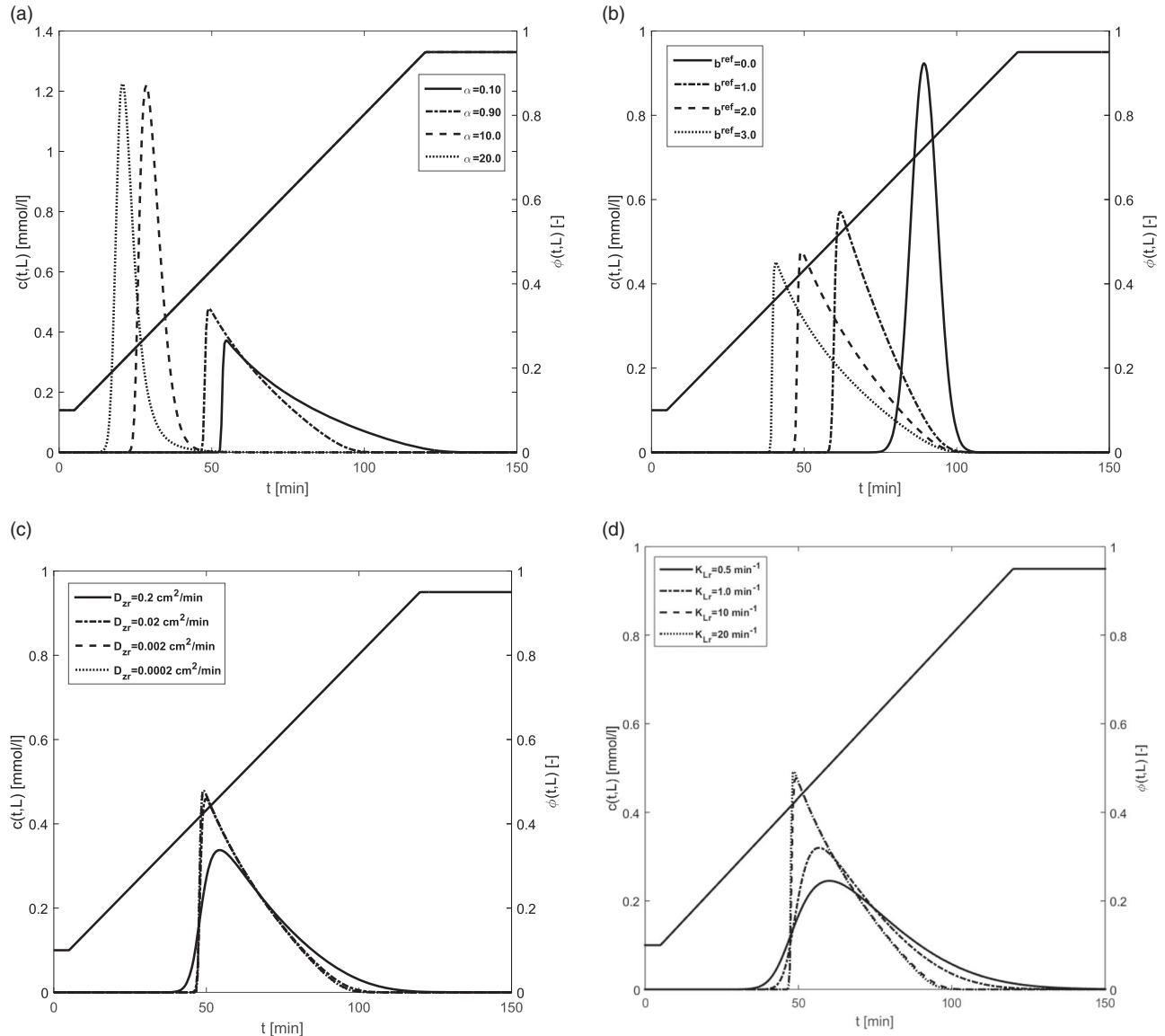


Figure 1. Effects of (a) component specific parameter α , (b) reference non linearity coefficient b^{ref} , (c) axial dispersion coefficient D_{zr} and (d) external mass transfer coefficient K_{Lr} on nonlinear single component elution profiles.

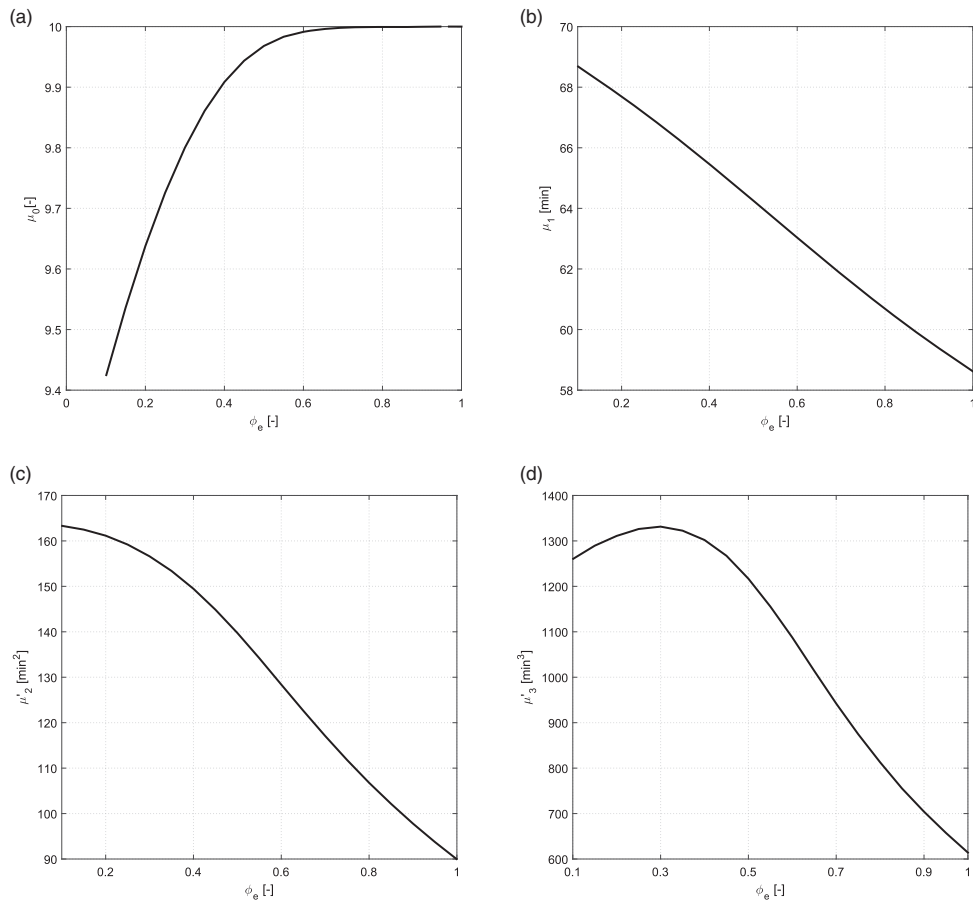


Figure 2. Effect of modulator concentration ϕ_e on temporal moments. ϕ_e has been changed gradually while other parameters are fixed as given in Table 1.

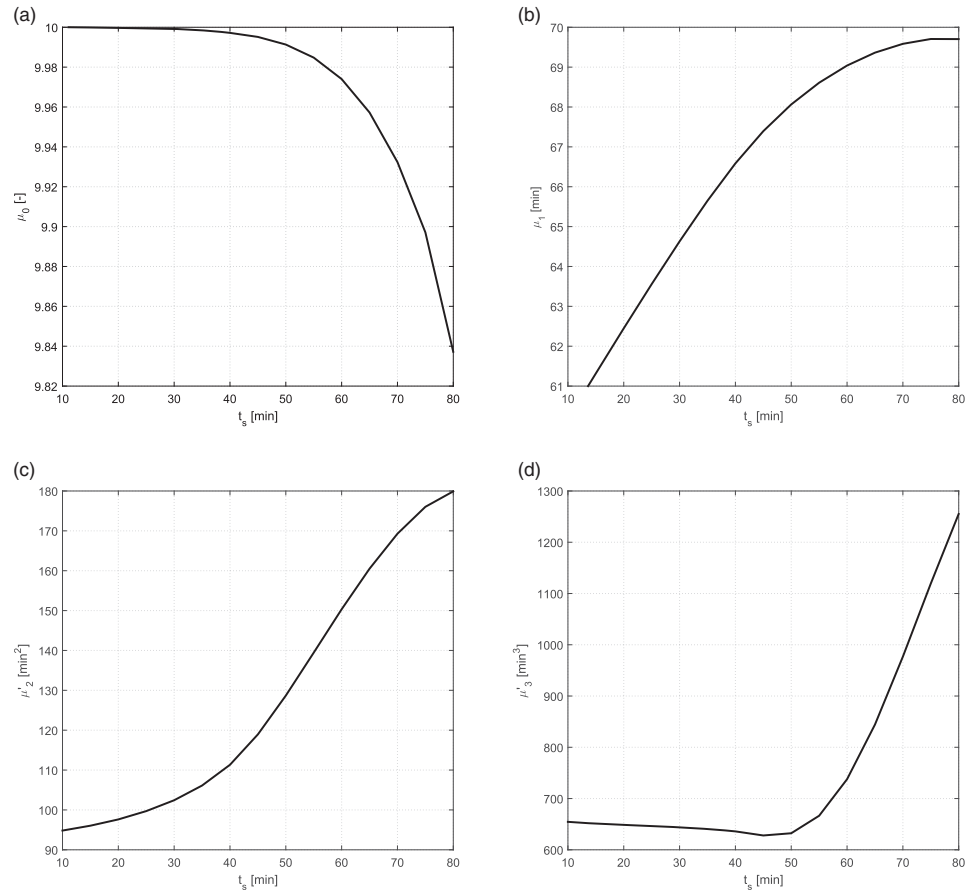


Figure 3. Effect of gradient start time t_s on temporal moments. t_s has been gradually changed while other parameters are fixed given in Table 1.

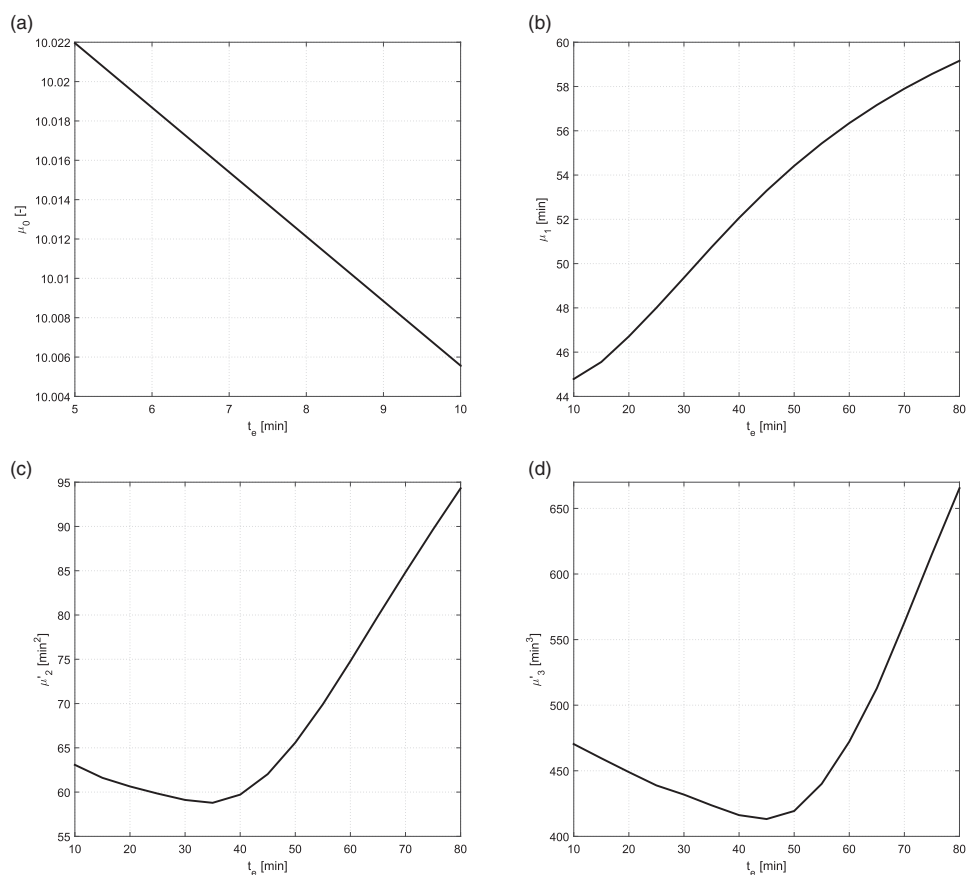


Figure 4. Effect of gradient end time t_g on temporal moments. t_g has been gradually changed while other parameters are fixed as given in Table 1.

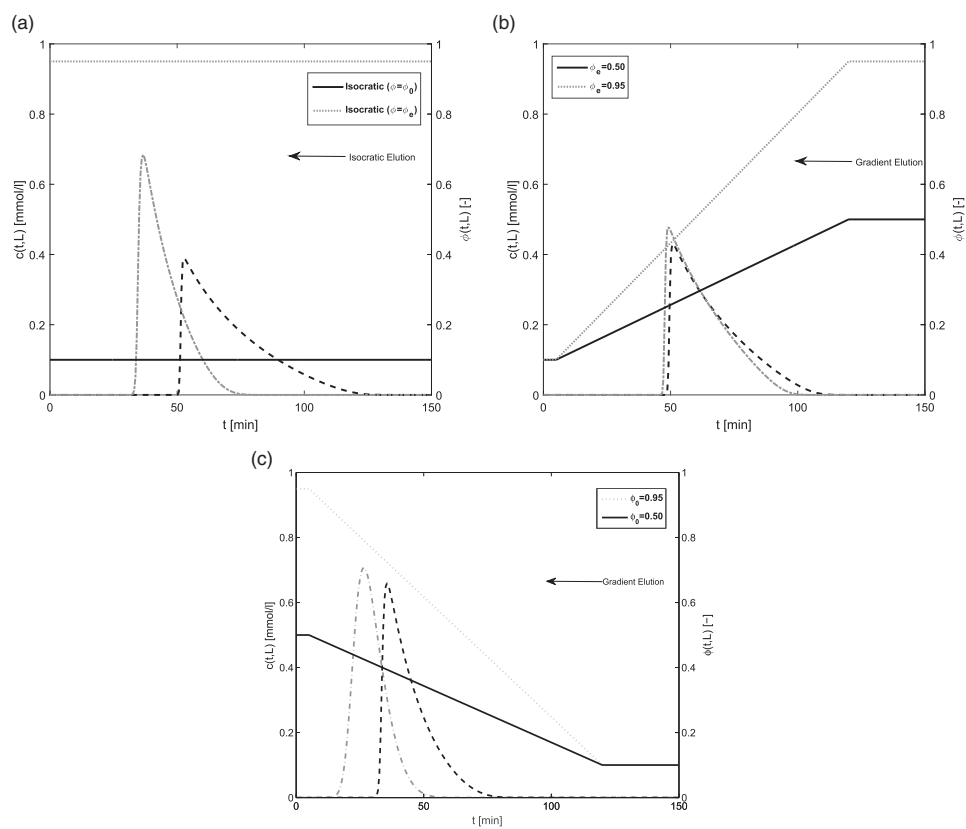


Figure 5. Comparison of isocratic and gradient elution for positive gradient and negative gradients respectively using single component elution. Other parameters are listed in Table 1.

third statistical moments are used. The peak area or total mass of the solute is estimated by the zero moment, the mean retention time of the profile in the column is estimated by first moment, while the second central moment provides information about spreading of the concentration profiles. Information about the asymmetry of the peak is obtained through the third central moment.

Numerical case studies

Here, the presented HR-FVM is applied to simulate different gradient elution chromatography processes. In the numerical test cases, single and two-component chromatography processes are considered. For the sake of simplicity, the reference axial dispersion and mass transfer coefficients are chosen the same, i.e. $D_{z,i} = D_{zr}$ and $K_{L,i} = K_{Lr}$, for $j = 1, 2, \dots, \mathcal{N}_c$. Furthermore, we have chosen the same value for all solvent strength parameters, i.e. $\alpha = \gamma = \zeta$. However, in practical applications they may differ and the current model equations allow such differences.

The single-component elution

At the current sub-section, few test cases are discussed for nonlinear single-component elution. We analyze the single-component 1-D nonlinear LKM given by Equations (1) to (4) with $N_c = 1$. The modulator concentration ϕ and the solute concentration c at the column outlet $z = L$ are plotted over time t in the figures. In addition, temporal moments are plotted in relation to various operational parameters in the model equations. Table 1 reflects the standard parameters utilized in the case studies. The parameters used in the test problems are set in accordance with ranges frequently found in HPLC applications.^[34]

Effects of α , b^{ref} , d_{zr} and K_{Lr} on nonlinear elution profiles

The influence of solvent strength parameter (α) on the nonlinear profiles is shown in Figure 1a. The retention time

Table 2. Parameters for both linear and nonlinear adsorption isotherms for two component case.

Parameters	Values
Column Length	$L = 10 \text{ cm}$
Porosity	$\epsilon = 0.4$
Velocity(Interstitial)	$u = 1.0 \text{ cm/min}$
Reference Henry's constant	$k_{Hr,1} = 5$
Reference Henry's constant	$k_{Hr,2} = 10$
Nonlinearity coefficient	$b_1^{ref} = 1$
Nonlinearity coefficient	$b_2^{ref} = 2$
Solvent strength parameter	$\alpha = 0.90$
Coefficient of mass transfer	$K_{Lr} = 10 \text{ min}^{-1}$
Injected concentration	$c_{inj,1} = 1 \text{ mol/l}$
Injected concentration	$c_{inj,2} = 1 \text{ mol/l}$
Injection time	$t_{inj} = 10 \text{ min}$
Initial concentration	$\phi_0 = 0.10$
Final concentration	$\phi_e = 0.95$
Starting time of Gradient	$t_s = 5 \text{ min}$
Ending time of Gradient	$t_e = 120 \text{ min}$
Initial concentration	$\phi_0 = 0.1$
Final concentration	$\phi_e = 0.95$

decreases as α changes from 0.1 to 1.0, i.e. as the sensitivity of Henry's constant to variation in modulator concentration increases. Figure 1b shows the effect of reference non-linearity coefficient b^{ref} on elution profile. For $b^{ref} = 0$ the elution profile has a gaussian shape. However, for increasing value of non-linearity coefficient, the retention time decreases and the elution peaks becomes more tailed on the right side. This is a typical Langmuir behavior normally observed in chromatographic processes. Figure 1c displays the effect of reference axial dispersion coefficient D_{zr} for different values using the non-linearity coefficient. The results shows that for decreasing value of D_{zr} the elution profile becomes more sharper. However, the mean retention remains unaffected. The effect of reference external mass transfer coefficient K_{Lr} , varying from $K_{Lr} = 1 \text{ min}^{-1}$ to 20 min^{-1} , using nonlinear isotherm can be seen in Figures 1d. The results show that elution peaks become more sharpened. However, as expected, the average retention time is not affected by mass transfer coefficient K_{Lr} . In other words, K_{Lr} has diffusion like effect on the elution profiles.

Effects of modulator concentration ϕ_e

Figure 2 displays the effect of gradually changing values of modulator concentration ϕ_e on the temporal moments. The first moment represents mean retention time, its value decreases with increasing the values of ϕ_e . The same behavior is observed for second and third central moment plots.

Effects of gradient ending time t_e

Figure 3 displays the moments plots to demonstrate the effects of starting time of gradient t_s for a fixed ending time t_e . The first moment represents mean retention time, its value increases with increasing values of t_s . The same behavior is observed for second and third central moment plots which represent the variance and asymmetry of elution profiles.

Effects of gradient ending time t_e

Figure 4 displays the effect of ending time of gradient t_e on moments with respect to fixed starting time t_s . The first moment represents mean retention time, its value increases with increasing values of t_e . The same behavior is observed for second and third central moment plots.

Comparison of isocratic and gradient elution for single-component case

Numerical simulation for the nonlinear single-component are given in Figure 5. The plots in Figure 5a shows that with $\phi = \phi_0$, the isocratic elution gives a wider elution profile with larger variance, larger retention time, and larger asymmetry. Whereas, the isocratic profile with $\phi = \phi_e$ produces a comparatively symmetric and narrower profile. Figure 5b shows that the profiles produced by positive gradient elution are narrower, symmetric, and asymmetric.

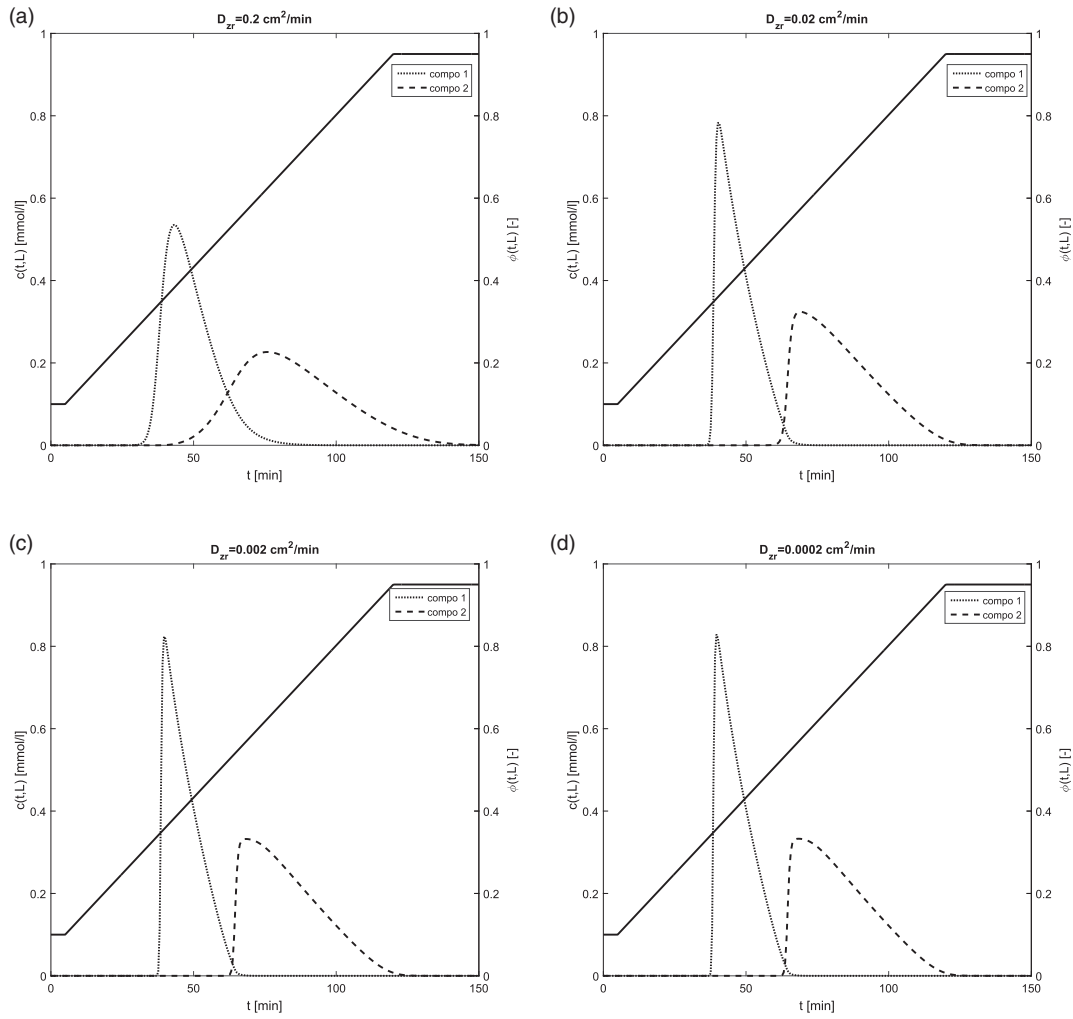


Figure 6. Effect of axial dispersion coefficient D_{zr} on elution profiles. D_{zr} has been gradually increased while other parameters are fixed given in Table 2.

In Figure 5c, the profiles generated by negative gradient elution are more symmetric and narrower than all other profiles.

The two-component elution

In this subsection, we have extended our current research analysis for the two-component nonlinear gradient elution. In these case studies, the numerical results are generated by solving the two-component 1D-KLM, given by Equation (1) and (2) for $N_c = 2$, with the HR-FVS of Koren.^[20,29] The non-linear langumir isotherms in Equation (3) are taken into consideration. The BCs given by Equations (10) and (11) for $N_c = 2$ are utilized to generate the numerical results. Once again the analysis is done by using the LSS model for the relationship between the component specific adsorption equilibrium constants, mass transfer, axial dispersion coefficient and the modulator concentrations. In the figures, the modulator concentration ϕ and the solute concentrations c_i are plotted at the column outlet $z = L$ as functions of the time t . Furthermore, influences of various operating parameters appearing in the model equations are depicted in the figures. In the case studies, the reference parameters used are specified in the Table 2.

Effects of d_{zr} on nonlinear two-component elution profiles

Figure 6 compares the numerical results generated for nonlinear elution profiles. A rectangular pulse of the liquid mixture containing the two components $c_{i, inj} = 1 \text{ mmol/l}$ for $i = 1, 2$ is infused in the flowing stream. The gradient parameters for this case studies have the values: $\phi_0 = 0.1$, $\phi_e = 0.95$, $t_s = 5 \text{ min}$, and $t_e = 80 \text{ min}$. In Figures 6, the numerical results for different values of axial dispersion coefficient D_{zr} are shown. It is depicted from the Figure 6a that the elution profiles become broaden along with elongated tails for $D_{zr} = 0.2 \text{ cm}^2 \text{ min}^{-1}$. Whereas, for other cases shown in Figure 6b for $D_{zr} = 0.02 \text{ cm}^2 \text{ min}^{-1}$, in Figure 6c for $D_{zr} = 0.002 \text{ cm}^2 \text{ min}^{-1}$ and in Figure 6d for $D_{zr} = 0.0002 \text{ cm}^2 \text{ min}^{-1}$ the elution profiles become more narrower having sharper peaks. Thus, showing that by decreasing the value of D_{zr} both components show improvement in separation and have sharper peaks. However, the mean retention remains unaffected.

Effects of α on nonlinear two-component elution profiles

Figure 7 shows the comparison of numerical results for different values of solvent strength parameter α . A rectangular

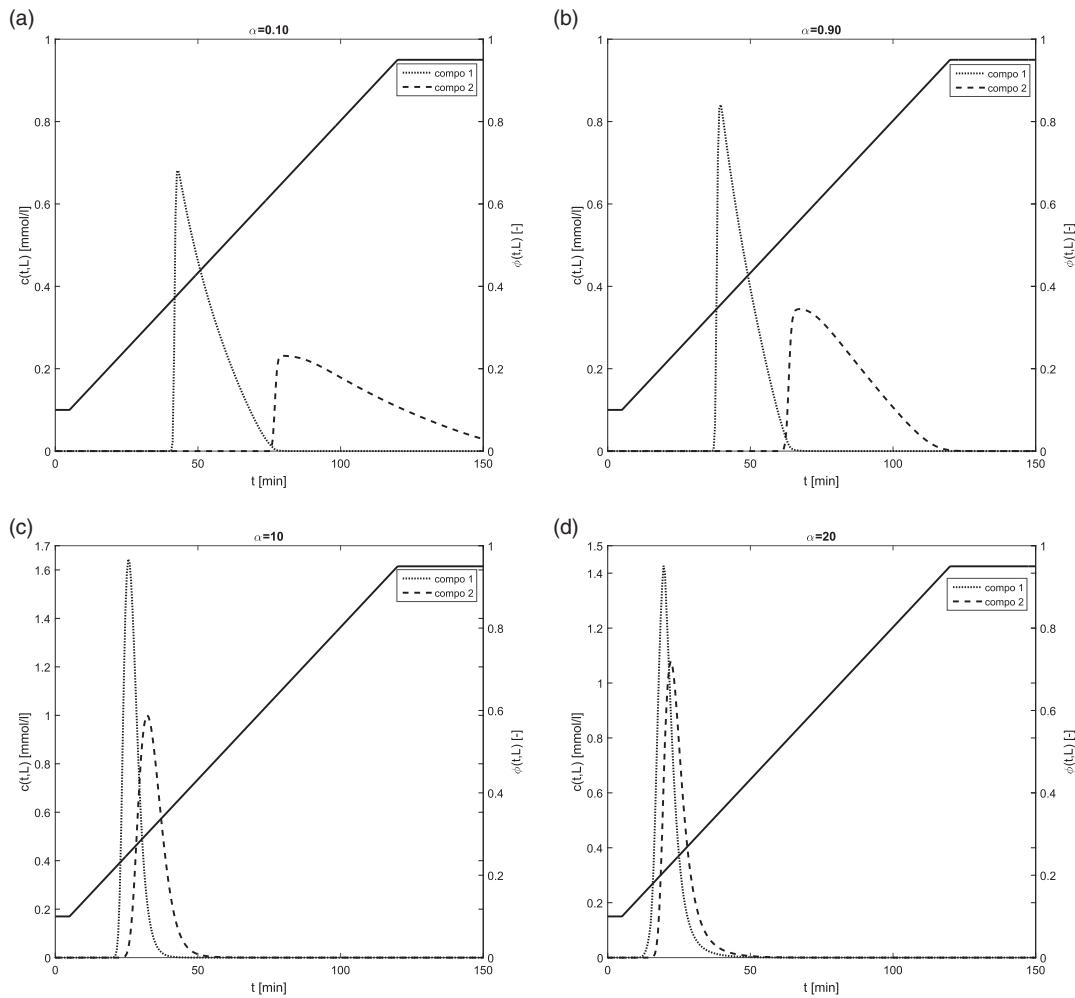


Figure 7. Effect of α the solvent strength parameter on elution profiles. α has been gradually increased while other parameters are fixed given in Table 2.

pulse of the liquid mixture containing the two components $c_{i, \text{inj}} = 1 \text{ mmol/l}$ for $i = 1, 2$ is infused in the flowing stream. The gradient parameters for this case studies are $\phi_0 = 0.1$, $\phi_e = 0.95$, $t_s = 5 \text{ min}$, and $t_e = 80 \text{ min}$. Figure 7a displays that the elution profiles become broaden along with elongated tail for $\alpha = 0.10$. Whereas, for increasing values of α shown in Figure 7b for $\alpha = 0.90$, in Figure 7c for $\alpha = 10$ and in Figure 7d for $\alpha = 20$ the elution profiles become more narrower having sharp fronts and asymmetrical peaks. By Increasing the value of α , improvement in the separation of components is noticed.

Comparison of isocratic and gradient elution for two-component case

Figure 8a presents numerical simulation for nonlinear two-component elution. The results show that same, like in the single-component case, the isocratic elution with $\phi = \phi_0$ creates wider profiles having larger variances, larger retention times, and larger asymmetries. The isocratic profile with $\phi = \phi_e$, however, creates symmetrical and narrow profiles. As demonstrated in Figure 8b, the profiles generated by gradient elution are more asymmetric and narrow than the other profiles. As displayed in Figure 8c, the profiles

generated by the negative gradient elution are most symmetric and narrow than the others.

In Figure 9, the outlet profiles of both components are plotted for isocratic and gradient elution. Here, we have chosen a case in which the value of Henry's constants differ significantly. The values of parameters used are: $D_{zr} = 0.002 \text{ cm}^2 \text{ min}^{-1}$, $b_1^{\text{bref}} = 1.0$, $b_2^{\text{bref}} = 2.0$, $k_{Hr,1} = 5 \text{ min}^{-1}$, $k_{Hr,2} = 20 \text{ min}^{-1}$, $t_s = 1 \text{ min}$, $t_e = 280 \text{ min}$, and other parameters are taken from Table 2. Due to rather different retention factors of the components, a wide gap is present between the elution profiles of both components during the isocratic elution. However, this gap is considerably minimizes during the gradient elution and their retention times also reduces. Larger or smaller gaps compared to the reference isocratic elution can result on changing the gradient steepness.

Conclusion

In this research paper, a nonlinear dispersive lumped kinetic model (LKM) of liquid chromatography was formulated to simulate the separation in overload columns using gradient elution chromatography. Nonlinear isotherm was investigated in which Henry constant and non-linearity coefficients depended on the solvent composition. Moreover, axial

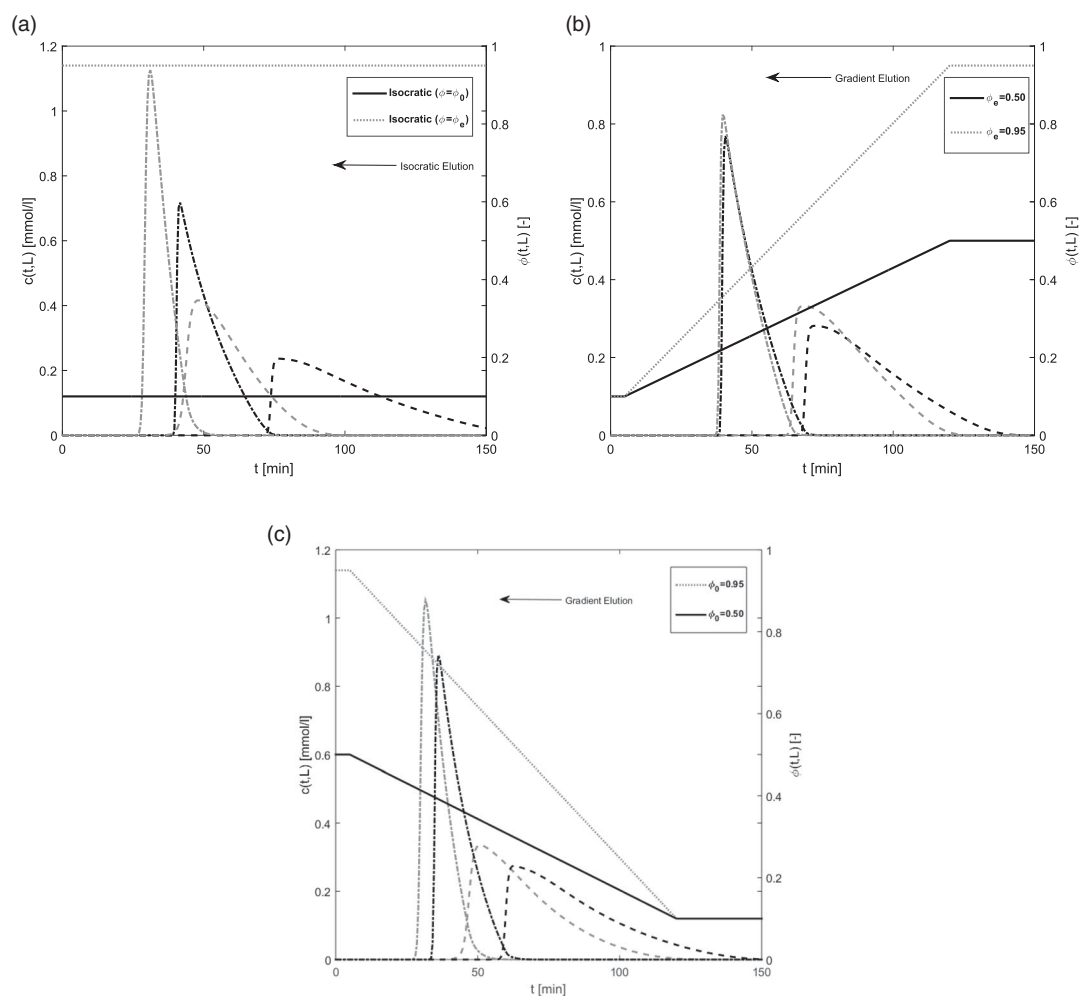


Figure 8. Plots (a), (b), and (c) are showing comparison of isocratic and gradient elution for positive and negative gradients using a mixture of two components. Other parameters are listed in Table 2.

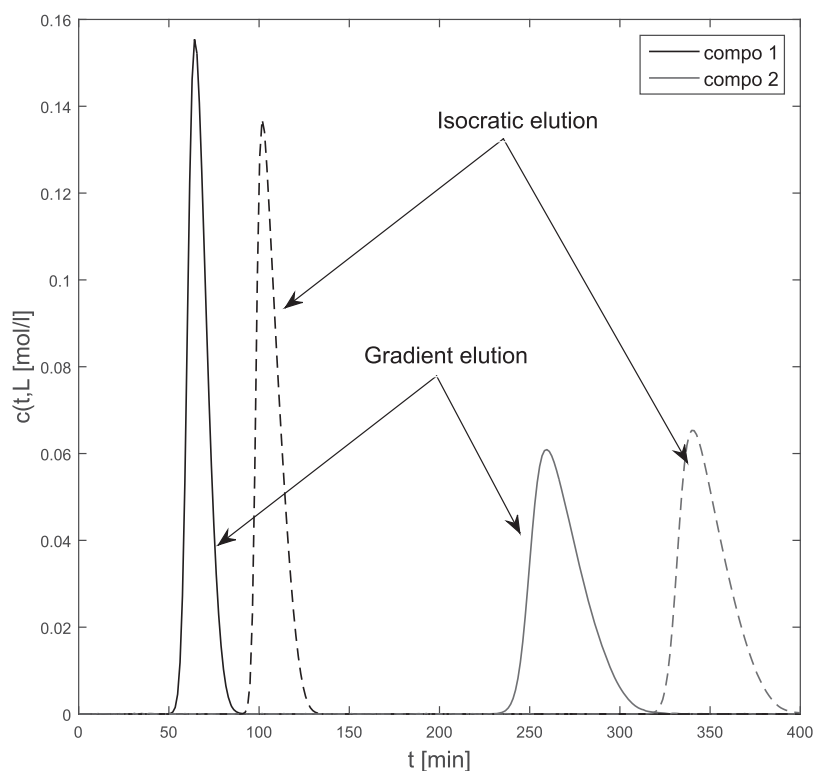


Figure 9. Plots showing comparison of isocratic and gradient elution for a mixture of two components having very different retention factors. Other parameters are listed in Table 2.

dispersion and mass transfer coefficients were also depending on the solvent composition. The suggested model was solved numerically by applying a high-resolution finite volume method considering an appropriate flux limiter. The method avoids numerical dissipation, resolve sharp discontinuities, and gives better accuracy. The influence of variation in modulator concentration was examined on the single and two-component elution. The influences of other parameters were also analyzed on the behavior of isolated components inside the column. For example, the scheme was utilized for studying the effects of gradient slopes, modulator concentration, Henry constant, non-linearity coefficients, axial dispersion coefficient, gradient start, end time and mass transfer coefficient on the concentration profiles. The proposed theoretical study will be very useful for optimizing the gradient elution chromatographic process.

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ORCID

Shamsul Qamar  <http://orcid.org/0000-0002-7358-6669>

References

- [1] Snyder, L. R.; Glajch, J. L.; Kirkland, J. J. *Practical HPLC Method Development*. 2nd ed.; Wiley-Interscience: New York, 1997.
- [2] Carta, G.; Stringfield, B. W. Analytic Solution for Volume-Overloaded Gradient Elution Chromatography. *J. Comput. Chem. Eng.* **1992**, *605*, 151–159. DOI: [10.1016/0021-9673\(92\)85232-I](https://doi.org/10.1016/0021-9673(92)85232-I).
- [3] Creasy, A.; Lomino, J.; Barker, G.; Khetan, A.; Carta, G. Gradient Elution Behavior of Proteins in Hydrophobic Interaction Chromatography with U-Shaped Retention Factor Curves. *J. Comput. Chem. Eng.* **2018**, *1547*, 53.
- [4] Jandera, P.; Churáček, J. *Gradient Elution in Column Liquid Chromatography*; Elsevier: Amsterdam, 1985.
- [5] Snyder, L. R.; Dolan, J. W. The Linear-Solvent-Strength Model of Gradient Elution. *Adv. Chromatogr.* **1998**, *38*, 115.
- [6] Jandera, P. Gradient Elution in Liquid Column Chromatography-Prediction of Retention and Optimization of Separation. *Adv. Chromatogr.* **2005**, *43*, 1–108.
- [7] Hao, W.; Zhang, X.; Hou, K. Analytical Solutions of the Ideal Model for Gradient Liquid Chromatography. *Anal. Chem.* **2006**, *78*, 7828–7840. DOI: [10.1021/ac061318y](https://doi.org/10.1021/ac061318y).
- [8] Hao, W.; Di, B.; Chen, Q.; Wang, J.; Yang, Y.; Sun, X. Study of the Peak Variance in Isocratic and Gradient Liquid Chromatography Using the Transport Model. *J. Chromatogr. A.* **2013**, *1295*, 67–81. DOI: [10.1016/j.chroma.2013.04.044](https://doi.org/10.1016/j.chroma.2013.04.044).
- [9] Lesins, V.; Ruckenstein, E. Gradient Flow Programming: A Coupling of Gradient Elution and Flow Programming. *J. Chromatogr.* **1989**, *467*, 1–14. DOI: [10.1016/S0021-9673\(01\)93947-0](https://doi.org/10.1016/S0021-9673(01)93947-0).
- [10] Zhu, P. L.; Dolan, J. W.; Snyder, L. R.; Hill, D. W.; Van Heukelem, L.; Waeghe, T. J. Combined Use of Temperature and Solvent Strength in Reversed-Phase Gradient Elution III. Selectivity for Ionizable Samples as a Function of Sample Type and pH. *J. Chromatogr. A.* **1996**, *756*, 51–62. DOI: [10.1016/S0021-9673\(96\)00723-6](https://doi.org/10.1016/S0021-9673(96)00723-6).
- [11] Viseras, C.; Cela, R.; Barroso, C. G.; Perez-Bustamante, J. A. Influence of Temperature in Reverse-Phase High-Performance Liquid Chromatography with Gradient Elution. *Anal. Chim. Acta.* **1987**, *196*, 115–122. DOI: [10.1016/S0003-2670\(00\)83076-7](https://doi.org/10.1016/S0003-2670(00)83076-7).
- [12] Van der Horst, A.; Schoenmakers, P. J. Comprehensive Two-Dimensional Liquid Chromatography of Polymers. *J. Chromatogr. A.* **2003**, *1000*, 693–709. DOI: [10.1016/S0021-9673\(03\)00495-3](https://doi.org/10.1016/S0021-9673(03)00495-3).
- [13] Snyder, L. R. Linear Elution Adsorption Chromatography: VII. gradient Elution Theory. *J. Chromatogr. A.* **1964**, *13*, 415–434. DOI: [10.1016/S0021-9673\(01\)95138-6](https://doi.org/10.1016/S0021-9673(01)95138-6).
- [14] Snyder, L. R. Principles of Gradient Elution. *Chromatogr. Rev.* **1965**, *7*, 1–51. DOI: [10.1016/0009-5907\(65\)80002-3](https://doi.org/10.1016/0009-5907(65)80002-3).
- [15] Jandera, P.; Churáček, J. Gradient Elution in Liquid Chromatography: I. The Influence of the Composition of the Mobile Phase on the Capacity Ratio (Retention Volume, Band Width, and Resolution) in Isocratic Elution — Theoretical Considerations. *J. Chromatogr.* **1974**, *91*, 207–221. DOI: [10.1016/S0021-9673\(01\)97901-4](https://doi.org/10.1016/S0021-9673(01)97901-4).
- [16] Snyder, L. R.; Dolan, J. W.; Gant, J. R. Gradient Elution in High-Performance Liquid Chromatography: I. Theoretical Basis for Reversed-Phase Systems. *J. Chromatogr.* **1979**, *165*, 3–30. DOI: [10.1016/S0021-9673\(00\)85726-X](https://doi.org/10.1016/S0021-9673(00)85726-X).
- [17] Guiochon, G.; Felinger, A.; Shirazi, D. G.; Katti, A. M. *Fundamentals of Preparative and Nonlinear Chromatography*. 2nd ed.; Elsevier Academic Press: New York, 2006.
- [18] Tingyue, G.; Yung-Huoy, T.; Gow-Jen, T.; George, T. Modelling of Gradient Elution Inmulti-Component Nonlinear Chromatography. *Chem. Eng. Sci.* **1992**, *47*, 253.
- [19] Miyabe, K.; Guiochon, G. Determination of the Lumped Mass Transfer Rate Coefficient by Frontal Analysis. *J. Chromatogr. A.* **2000**, *890*, 211–223. DOI: [10.1016/S0021-9673\(00\)00537-9](https://doi.org/10.1016/S0021-9673(00)00537-9).
- [20] Javeed, S.; Qamar, S.; Seidel-Morgenstern, A.; Warnecke, G. Efficient and Accurate Numerical Simulation of Nonlinear Chromatographic Processes. *J. Comput. Chem. Eng.* **2011**, *35*, 2294–2305. DOI: [10.1016/j.compchemeng.2010.10.002](https://doi.org/10.1016/j.compchemeng.2010.10.002).
- [21] Kaczmariski, K.; Antos, D. Fast Finite Difference Method for Solving Multicomponent Adsorption-Chromatography Models. *Comput. Chem. Eng.* **1996**, *20*, 1271–1276. DOI: [10.1016/0098-1354\(95\)00247-2](https://doi.org/10.1016/0098-1354(95)00247-2).
- [22] Coimbra, M. C.; Sereno, C.; Rodrigues, A. Modelling Multicomponent Adsorption Process by a Moving Finite Element Method. *J. Comput. Appl. Math.* **2000**, *115*, 169–179. DOI: [10.1016/S0377-0427\(99\)00171-5](https://doi.org/10.1016/S0377-0427(99)00171-5).
- [23] Javeed, S.; Qamar, S.; Seidel-Morgenstern, A.; Warnecke, G. A Discontinuous Galerkin Method to Solve Chromatographic Models. *J. Chroma. A.* **2011**, *1218*, 7137–7146. DOI: [10.1016/j.chroma.2011.08.005](https://doi.org/10.1016/j.chroma.2011.08.005).
- [24] Püttmann, P. A.; Nicolai, M.; Behr, M.; von Lieres, E. Stabilized Space-Time Finite Elements for High-Definition Simulation of Packed Bed Chromatography. *Finite Elem. Anal. Des.* **2014**, *86*, 1–11. DOI: [10.1016/j.finela.2014.03.001](https://doi.org/10.1016/j.finela.2014.03.001).
- [25] Cruz, P.; Santos, J. C.; Magalhães, F. D.; Mendes, A. Simulation of Separation Processes Using Finite Volume Method. *J. Comput. & Chem. Eng.* **2005**, *30*, 83–98. DOI: [10.1016/j.compchemeng.2005.08.004](https://doi.org/10.1016/j.compchemeng.2005.08.004).
- [26] Webley, P. A.; He, J. Fast Solution-Adaptive Finite Volume Method for PSA/VSA Cycle Simulation; 1 Single Step Simulation. *J. Comput. Chem. Eng.* **2000**, *23*, 1701–1712. DOI: [10.1016/S0098-1354\(99\)00320-8](https://doi.org/10.1016/S0098-1354(99)00320-8).
- [27] Leveque, R. J. *Numerical Methods for Conservation Laws*; Birkhauser Verlag: Basel, Germany, 1992.
- [28] Leer, B. V. Towards Ultimate Conservative Finite Difference Scheme.IV. A New Approach to Numerical Convection Scheme. *J. Comput. Phys.* **1997**, *23*, 276.
- [29] Koren, B. A. Robust Upwind Discretization Method for Advection, Diffusion and Source Terms, Notes on Numerical

- Fluid Mechanics. *Numer. Method Advect Diff Problem*. **1993**, 45, 117.
- [30] Qamar, S.; Rehman, N.; Carta, S.; Seidel-Morgenstern, A. Analysis of Gradient Elution Chromatography Using the Transport Model. *Chem. Eng Sci.* **2020**, 225, 115809. <https://doi.org/10.1016/j.ces.2020.115809>
- [31] Ruthven, D. M. *Principles of Adsorption and Adsorption Processes*; John Wiley and Sons, Wiley-Interscience: New York, 1984.
- [32] Dondi, F.; Guiochon, G. *Theoretical Advancement in Chromatography and Related Separation Techniques*; Springer Science + Business Media Dordrecht Ferrara: Italy, 1992.
- [33] Ortner, F.; Jermann, S.; Joss, L.; Mazzotti, M. Equilibrium Theory Analysis of a Binary Chromatographic System Subject to a Mixed Generalized Bi-Langmuir Isother. *Ind. Eng. Chem. Res.* **2015**, 54, 11420–11437. DOI: [10.1021/acs.iecr.5b03089](https://doi.org/10.1021/acs.iecr.5b03089).
- [34] Gritti, F.; Guiochon, G. Analytical Solution of the Ideal Model of Chromatography for a Bi-Langmuir Adsorption Isotherm. *Anal. Chem.* **2013**, 85, 8552–8558. DOI: [10.1021/ac4015897](https://doi.org/10.1021/ac4015897).
- [35] Demin, A. A.; Melenevsky, T. A. Bi-Langmuir Isotherms' Applicability for Description of Interaction of Ion-Exchange Sorbents with Protein Mixtures. *J. Chromatogr. Sci.* **2006**, 44, 181–186. DOI: [10.1093/chromsci/44.4.181](https://doi.org/10.1093/chromsci/44.4.181).
- [36] Sweby, P. K. High Resolution Schemes Using Flux Limiters for Hyperbolic Conservation Laws. *SIAM J. Numer. Anal.* **1984**, 21, 995–1011. DOI: [10.1137/0721062](https://doi.org/10.1137/0721062).